Veins

Veins are blood vessels that carry blood towards the heart. Blood vessels receive blood from the venules and transport blood back to the heart. Like arteries, veins, have three layers.

Blood Components

Blood is a body fluid that consists of:

- Plasma
- Red Blood Cells
- White Blood Cells
- Platelets

Plasma

Plasma in blood is over 50% of the volume of blood and over 90% of plasma is water. The main component of plasma is plasma albumin which is a protein that enables and controls the osmotic pressure of the blood.

Red Blood Cells (RBC)

Red blood cells or erythrocytes are disc like in shape. RBCs are enucleated, do not contain a nucleus. The red blood cells contain iron laden haemoglobin which carries oxygen to the cells. Red blood cells also contain glycoproteins which determine the blood group of an individual. The blood types are type A, type B, type AB and type O.

White Blood Cells

White blood cells or leukocytes are part of the immune system which fights against infections from pathogens. When the white blood cell count rises, it is a sign of infection. All leukocytes have a distinct nucleus.

The various types of white blood cells are:

- Eosinophils
- Basophils
- Neutrophils
- Lymphocytes
- Monocytes

Platelets

Platelets or thrombocytes(do not have a nucleus) maintain haemostasis (clotting). Haemostasis is enabled by the coagulation or thickening of blood by production of fibrin from the clotting factors found within the platelets to prevent blood loss, when a blood vessel has been broken due to injury.

Lymphatic system

The lymphatic subsystem, a part of the circulatory system, is closely aligned with the body's immune system and removes excessive fluid from the body. The lymphatic vessels contain lymph including lymphocytes. Lymphatic system also consists of lymph nodes and lymphatic organs. These organs include the **thymus gland**, **spleen**, **bone marrow** and **tonsils**. Nodes are found throughout the lymphatic system and they serve to filter the blood as it travels throughout the body. **Swollen lymph nodes** are a signal of a disease or an **infection**. Many lymph nodes are found in the neck area, under the arms and in the groin area, although there are hundreds of them throughout the body.

Spleen

Spleen is an organ present in the upper left part of the abdomen and to the left side of the stomach. The spleen plays multiple supporting roles in the body. It acts as a filter for blood as part of the immune system. Old red blood cells are recycled in the spleen, platelets and white bloods are stored.

Diseases related to blood and blood vessels

- Phlebitis
- Deep Vein Thrombosis (DVT)
- Anaemia, including pernicious anaemia and sickle cell anaemia
- Leukaemia
- Lymphoma
- Thrombocytopenia.

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